Hernan Dario Mojica Diaz

13hernanm@gmail.com | Curriculum Vitae, 22nd October 2026

PROFESSIONAL PROFILE

I am a mechanical engineer and have directed my professional career toward two main areas.
The first is building products and companies: doing hard things and sustaining the resilience that process demands. That means developing technology, talking to customers, and iterating until the solutions connect with people's real needs. Here I develop software and *AI First* projects, including agent programming, machine learning architectures, process automation, and systems integration.
The second is active involvement in applied research, particularly in energy, climate change, data analysis, and systems modeling.
My recent work centers on two projects: OrionAtiende.com, a SaaS of AI agents for sales and customer service over WhatsApp, and SatEmis, a climate research platform that applies machine learning to satellite data at scale. Orion has been my most challenging project: it has taught me to build a product, sell it, face rejection, manage resources, and iterate, making decisions based on data rather than feelings.
As a mechanical engineering consultant, I work in thermal and fluid systems: renewable energy, energy modeling, computational fluid dynamics (CFD), climate change mitigation, emissions inventories, and remote monitoring.
I have nearly four years of certified experience as a lead engineer in software design and development, building solutions that integrate data, artificial intelligence, and applied engineering.

EDUCATION

National University of Colombia, Bogotá, Colombia *February 2018 - December 2024*
Degree in *Mechanical Engineering*
Thesis Title: *Fluid-dynamic analysis of a propulsion mini-turbine*
Supervisor: Dr. Juan Miguel Mantilla

SKILLS

Spoken Languages:
English B2 (Professional), Spanish (Native)
Programming Languages, Tools and Technologies:
Python, JavaScript, Git, Linux, Google Cloud Platform (GCP), nest.js, Docker, Pytorch
LLMs and AI Tools:
Deep Learning, Machine Learning, Databases (SQL, NoSQL and vector), MCP, LlamaIndex, Ollama
Engineering Tools and Technologies:
TIMES Model, Autodesk Software, Microsoft Office, Ansys, OpenFoam
Soft Skills:
Teamwork, Leadership, Problem Solving, Communication, Adaptability

PROFESSIONAL EXPERIENCE

Orion AI *March 2026 - Present*
Founder and Full-Stack Developer

- Founded and built Orion AI (orionatiende.com) end to end: a multi-tenant SaaS in production that handles WhatsApp conversations with AI agents and turns each one into an actionable CRM. I own the product, architecture, billing, and go-to-market.

- Developed the backend with FastAPI and PostgreSQL and the frontend with Next.js 16 (React, TypeScript, Server Components, and a BFF). It includes per-tenant data isolation, asynchronous workers with an outbox and idempotent retries, and nearly 1,900 automated tests across almost 600 commits.
- Designed the conversational engine with LangGraph: persistent memory, voice note transcription, and tools to query knowledge, move deals through the funnel, and hand off to human agents in Chatwoot. Contacts and deals are created deterministically before the model is invoked, so the LLM only modifies records that already exist.
- Built a RAG system on Qdrant for PDF, Markdown, and TXT, with configurable embeddings, agent-isolated search, and a durable indexing worker. Also integrated a classifier model that places each conversation in its funnel stage.
- Drove the Meta integration to completion: business verification, approved App Review for the WhatsApp Business permissions, and an in-house Embedded Signup v4 flow that lets each company connect its number without manual support.
- Implemented recurring subscriptions with Wompi (cards with 3DS, renewals, retention, and plan-based access control) and platform security: passwordless login (email code and Google), JWT, fine-grained RBAC, credential encryption, and rate limiting. I run the system on Railway and a VPS with Docker and self-hosted Chatwoot, with Prometheus metrics and Grafana alerts.
- Validated the market in the field by calling +200 tourism businesses (70 responses and 30 demos), and currently have 8 paying clients. Built a prospecting pipeline on open data from Colombia's National Tourism Registry (about 95,000 businesses) and a Meta Ads catalog for an iterative experiment to discover the ideal customer profile.
- Organized development as a team of AI agents (Claude Code and OpenCode), one session per repository, coordinated through a knowledge base that an LLM pipeline keeps in sync with the commits.

SatEmis *August 2025 - Present*

Software and Artificial Intelligence Engineer

- Designed and implemented a distributed architecture with NestJS, FastAPI, Docker, and PostgreSQL/PostGIS to manage emissions inventories and geospatial processing.
- Developed a satellite data service with Google Earth Engine and Sentinel-5P to query, aggregate, and analyze methane concentrations by department in Colombia.
- Built APIs for the ingestion, processing, and retrieval of climate inventories and geospatial data.
- Developed a React and TypeScript web application integrating authentication, interactive maps, emissions visualization, and internationalization.
- Integrated artificial intelligence capabilities to retrieve and contextualize climate regulations, facilitating the interpretation of environmental information.
- Containerized and optimized services on Google Cloud Platform (GCP) to improve processing and storage resource utilization.

National University of Colombia *February 2024 - July 2025*

Energy Systems Researcher and Analyst

- Development of Colombia's national energy reference model using the TIMES VEDA framework (IEA-ETSAP).
- Collection and processing of Colombian energy system data by economic sectors and regions.
- Processing of large datasets and development of analytical tools using Python.
- Projection of energy transition scenarios aligned with national climate change plans.
- Leadership and coordination of multidisciplinary teams.
- Effective communication of technical results to non-technical audiences.

National University of Colombia *April 2022 - February 2024*
Research Assistant - Faculty of Engineering

- Support in developing the mitigation section of the Comprehensive Climate Change Plan (PIGCC) for power sector companies participating in the Carbon-Neutral Power Sector Alliance (ASECN).
- Quantification of Scope 1, 2 and 3 carbon footprints using the GHG Protocol methodology.
- Development of statistical models to project emissions based on operational variables.
- Assessment of mitigation measures such as energy efficiency, fuel switching, changes in the energy mix and carbon offsetting.
- Statistical studies on the relationship between climate phenomena and power generation by source.
- Preparation of presentations for stakeholder engagement and dissemination of results.

TEACHING AND VOLUNTEERING

National University of Colombia *February 2023 - November 2023*
Instructor Supervisor: M.Sc Luis Carlos Trujillo
Department of Human Sciences
Bogotá, Colombia

- Teaching introductory programming with Python to students of the Faculty of Human Sciences.
- Design of a teaching strategy adapted to non-engineering students.
- Creation of educational presentations to improve knowledge transfer.
- Introduction of basic Git concepts for collaborative project management.

CERTIFICATIONS

Introduction to Numerical Analysis *Graduated November 2020*
Coursera

View Certificate

Process Engineering for the Energy Transition *Graduated July 2023*
National University of Colombia

View Certificate

MCP: Build Rich-Context AI Apps *October 2025*
DeepLearning.AI

View Certificate

Sustainable Supply Chain Management *Graduated December 2025*
MITx

View Certificate

Machine Learning with Python: from Linear Models to Deep Learning *Graduated December 2025*
MITx

View Certificate

PUBLICATIONS

Juan Miguel Mantilla González Julián Esteban Olejua Pinto, *Analysis of Methane Sources and Sinks in Colombia Using Copernicus S5P Data*, En publicación, 2025 [Link].